

Power-aware test scheduling framework for IEEE 1687 multi-power domain networks using formal techniques[☆]

Payam Habiby^{a,*}, Sebastian Huhn^{a,b}, Rolf Drechsler^{a,b}

^a Cyber-Physical Systems, DFKI GmbH, 28359 Bremen, Germany

^b Institute of Computer Science, University of Bremen, 28359 Bremen, Germany

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ABSTRACT

The IEEE 1687 Std. (IJTAG) introduces an efficient access methodology based on reconfigurable scan networks to address the ever-increasing complexity of the latest system-on-chips. By invoking this new methodology, the overall test time can considerably be reduced by shortening the scan chains' length without sacrificing the test quality. IJTAG allows for designing highly complex test networks compromising the latest test structure that enables to achieve the high test quality, as required by today's customers' applications. Besides the time overhead induced by the required network (re-)configuration, the access sequence of the instruments itself greatly affects the overall test time. Furthermore, the power characteristics of the complex test facilities have to be taken into account to avoid effects like IR drop during the later test application that is even more critical in highly multi-power domain networks. Consequently, the IJTAG methodology strictly requires an effective test scheduler that considers the individual instruments' constraints and is compatible with recent multi-power domain chip designs. This work proposes two novel power-aware test scheduling approaches based on pseudo-Boolean optimization and integer linear programming techniques that are both seamlessly integrated into a fully-automated test scheduling framework for IJTAG test networks even with multiple power domains. The first proposed optimization scheme allows for determining a local optimal test schedule, applicable to networks with more than one thousand instruments since the required run-time is manageable. Furthermore, the second optimization scheme determines a global optimal test scheduler, which is most suitable for mid-sized networks in which it is clearly outperforming any other existing technique.

1. Introduction

The state-of-the-art *System-on-Chips* (SoCs) embrace a large number of heterogeneous embedded components like digital-signal processors, sensors, communication and interface modules. The steadily increasing complexity of these chips has created a growing demand for effective infrastructures that enable efficient access methodologies for test, debug, and monitoring. Following this demand, IEEE 1687 Std. (IJTAG) has introduced new reconfigurable scan networks to tackle the challenge of long scan chains by adding programmable components such as *Segment Insertion Bits* (SIBs) and *ScanMux Control Bits* (SCBs) to the IJTAG test network (1). Despite the extensive test capabilities provided by this reconfigurable scan network, further technical challenges like IR drop (2,3) have to be addressed since, otherwise, the parallel test of all concurrently accessible instruments is impossible. The higher

complexity of this problem in multi-power domain SOC's emphasizes the urgent need for more effective access management.

In an IJTAG network, any on-chip module that holds only a client interface and is accessed by an external controller, is referred to as an *instrument*. The required configuration data for SIBs and SCBs is transferred serially along with the test data during every *Capture-Shift-Update* (CSU) cycle. In every CSU cycle, all instruments of the current active chain receive the new data from *Scan-In* (SI) port and at the same time shift their individual data to *Scan-Out* (SO) port of the chip.

In order to eliminate the time penalty due to excessive configuration data, SIBs enable the creation of hierarchical scan chain designs. More precisely, SIBs can include or exclude a segment of the network that is connected to their host interface. This provides more efficient access to different parts of the SoC in comparison to the legacy daisy scan chains. Subsequently, a faster instrument access mechanism significantly

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* Corresponding author.

E-mail addresses: payam.habiby@dfki.de (P. Habiby), huhn@uni-bremen.de (S. Huhn), drechsler@uni-bremen.de (R. Drechsler).

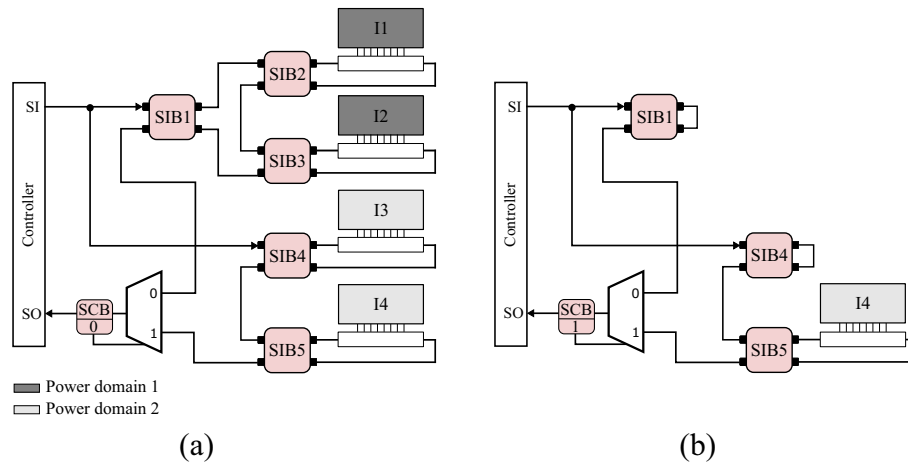


Fig. 1. Reconfigurable instrument access in IJTAG network.

reduces the test cost since these costs are strongly affected by the overall test time. This concept is further described in (4) by introducing a method for calculating the overall instrument access time.

This paper heavily exploits formal techniques to develop a power-aware fully-automated test scheduling framework for IEEE 1687 multi-power domain networks. At first, a *Boolean Satisfiability* (SAT)-based model is presented, which is applicable for arbitrary IJTAG test network architectures comprising SIBs, ScanMuxes, and SCBs. The optimization of test scheduling by considering multiple power domains is implemented by employing incremental *Pseudo-Boolean Optimization* (PBO) techniques. Various experiments clearly prove that this approach can be applied to networks containing over one thousand instruments without any concerns regarding the resulting CPU run-time. On top of this SAT-based approach, a novel *Integer Linear Programming* (ILP) model is proposed, which allows determining even global optimal test sequences for IJTAG test networks. The proposed models are processed by formal solving engines yielding power-safe tests and, hence, allowing for a significant test cost reduction.

The remainder of this paper discusses the structure and evaluation of the proposed framework as follows: Section 2 briefly reviews related works regarding the instrument access in IJTAG networks. Section 3 provides a comprehensive description of the background that is needed to make this paper self-contained – including a review on SAT- and ILP-based techniques and IJTAG test networks. Afterwards, the general idea and the general flow of the proposed test schedulers are described in Section 4. The conducted experimental results are summarized and discussed in Section 5. Finally, conclusions are drawn in Section 6.

2. Related work

Several techniques have been proposed to improve different aspects of the IJTAG networks (5–17). A hybrid scheduling scheme is presented in (6) to create a purely SIB-based network with minimized SIB programming overhead. A new structure for SIB is proposed in (7) to increase the bandwidth of the IJTAG network. This idea is further enhanced in (8) to create a scalable broadcast design that reduces the test time by sending identical data to the replicated cores. Other works like (11,18) have improved the access time without applying any structural modifications. In (9,19), the authors have exploited the Boolean satisfiability problem to optimize the instrument access time for scan pattern retargeting in reconfigurable scan networks that do not include SIBs. This approach has been further improved by introducing an upper bound calculation method in (11).

Due to the power, temperature, and other resource limitation, the simultaneous test of all concurrently accessible instruments is not feasible, and therefore the use of an instrument access scheduling

scheme is inevitable. Optimization of the test time by using power constrained scheduling is investigated in (5,12,13). A session-less test schedule about sorting the instrument based on the number of test patterns and resource conflicts is introduced in (12). The method proposed in (13) uses a modified multi-bit SIB to follow a parallel scheduling approach based on an exhaustive search over all possible solutions. Typically, the embedded cores in the network require to be accessed a different number of times. The sequence and number of access in each test session considerably affect the overall test duration and, hence, an effective test scheduler has a major contribution to the reduction of final test cost.

Generally, modern SoCs are divided into multiple power domains, which can be powered on or off selectively to manage the total power dissipation and avoid overheating (20). This important feature has not been considered in any test scheduler for IJTAG networks in the literature so far. However, for modern test schedulers, it is essential to consider the resulting power characteristics of the resulting test set, as being addressed by this work. Although IJTAG provides more flexible access to the instruments, the implementation of multiple power domains along with the hierarchical structure of the network, non-linear constraints, and the existence of different concurrency groups as discussed in (14) poses new challenges against test scheduling. In work (14), a graph-based model of IJTAG networks has been proposed to detect the concurrently accessible instruments in multi-power domain networks. The access and power constraints are exploited to prune the search tree during the exhaustive search based on dynamic programming. Despite the promising results, using this method in large networks requires intensive computations.

In contrast to any other existing related work, this paper proposes a power-aware fully-automated test scheduler framework for IJTAG multi-power domain networks by heavily exploiting formal techniques while considering the instrument-specific power constraints to, in the end, determine highly effective but power-safe tests.

3. Background

This section presents the required preliminaries for the comprehension of this work. In particular, a brief introduction to IJTAG networks and the formal techniques – like the SAT problem and ILP-based modeling – are given.

3.1. IJTAG networks

The basic structure of IJTAG networks and their reconfiguration mechanism is described in this section. Fig. 1 presents an exemplary IJTAG network, which provides reconfigurable access to the instruments

I_1 to I_4 . These instruments are connected to the network through parallel *Test Data Registers* (TDRs), which are divided into two power domains indicated by different colors. The test data is shifted to the TDRs serially and then transferred concurrently via parallel data ports to the instruments. The ScanMux sensitizes one of two branches to create a chain between SI and SO. SIBs operate as 1-bit programmable switches that are closed by resetting them to 0 as soon as their client segment does not require further access. By this, a shorter path is provided for accessing the rest of the network's instruments.

In Fig. 1(a), it is assumed that I_1 and I_2 are both accessible in the initial state by setting the SCB to 0 and SIB_1 , SIB_2 , and SIB_3 to 1. In other words, the active chain is $\{SI \rightarrow SIB_1 \rightarrow SIB_2 \rightarrow I_1 \rightarrow SIB_3 \rightarrow I_2 \rightarrow SCB \rightarrow SO\}$. Assuming that only I_4 is required to be accessed in the next session, changing the value of SCB to 1 and setting SIB_4 and SIB_5 to 0 and 1 creates the new active chain $\{SI \rightarrow SIB_4 \rightarrow SIB_5 \rightarrow I_4 \rightarrow SCB \rightarrow SO\}$, as shown in Fig. 1(b).¹ The configuration data for the next session is always concatenated into the current test data and shifted through the active chain. During the capture operation, the data is loaded to TDRs from the corresponding instruments. In the next phase, the content of all TDRs is transferred to the SO port and the new data is simultaneously shifted from SI in. Finally, the instruments receive the new data through an update signal.

The IJTAG standard introduces two languages that are used to build up the underlying network structure and to reflect the required test data retargeting. First, the *Instrument Connectivity Language* (ICL) that describes the network structure on an abstract level without revealing the functional logic of the design. Secondly, the *Procedural Description Language* (PDL) defining the sequence of operations, as required for accessing an instrument. In this paper, the ICL netlist is utilized to create a Boolean network description, which is subsequently used to analyze the network for obtaining an optimized access schedule.

3.2. Boolean satisfiability

The SAT problem attempts to determine whether a set of variable assignments of a given Boolean function exists such that the function is satisfied, i.e., is evaluated to *TRUE*. If such an assignment exists, the function is considered to be *satisfiable* (sat) and otherwise *unsatisfiable* (unsat) (21). A lot of research work has been spent on the development of powerful SAT solving algorithms yielding well-engineered SAT solvers like clasp (22), MiniSAT (23), SLIME (24), and ManySAT (25). This is since the SAT problem has recently been utilized to address various research questions effectively in domains like the structural test generation (26).

In the Boolean algebra, a literal $\hat{x}$ is either a Boolean variable (x) or its negation ($\bar{x}$). A disjunction of literals $\bigvee_{i=1}^n \hat{x}_i$ forms a clause. A SAT instance is a Boolean function $\Phi: \mathbb{B}^n \rightarrow \mathbb{B}$ where $\mathbb{B}$ is the set of Boolean values 0 and 1. Φ is a conjunction of clauses and is typically represented as *Conjunctive Normal Form* (CNF). An exemplary SAT instance is presented in Example 1.

Example 1. Let $\Phi = (\bar{x}_1 \vee x_2 \vee x_3) \cdot (x_1 \vee x_3) \cdot (\bar{x}_3)$. Then, $x_1=1$, $x_2=1$, $x_3=0$ is a satisfying assignment.

A Boolean function $\Psi: \mathbb{B}^n \rightarrow \mathbb{R}$ with weighted literals is called a *Pseudo-Boolean* (PB) function and the corresponding PB-SAT problem is defined as follows:

$$\sum_{i=1}^{n-1} w_i \cdot \hat{x}_i \geq w_n \text{ with } w_i \in \mathbb{R} \quad (1)$$

This expression is used as a constraint in PBO problems. PBO incorporates an objective function $\mathcal{F}$ during the solving process such that

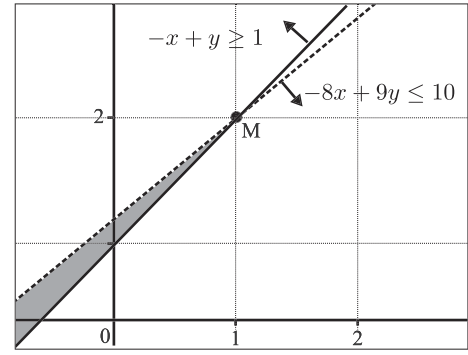


Fig. 2. Integer linear programming example.

a solution does not only satisfies the constraints but also optimizes the solution with respect to a given $\mathcal{F}$. Such an objective function is typically expressed as the arithmetic sum of all PB co-factors as follows:

$$\mathcal{F}(x_1, \dots, x_k) = \sum_{i=1}^k w_i \cdot \hat{x}_i \text{ with } w_i \in \mathbb{R} \quad (2)$$

An exemplary PBO instance is given in Example 2.

Example 2. Let $\Psi = (\bar{x}_1 + 2x_2 + 3x_3 \geq 3) \cdot (2\bar{x}_1 + 3x_2 > 4)$ and $\mathcal{F} = x_1 + x_2 + x_3$. Then, $x_1=0$, $x_2=1$, $x_3=0$ is a solution to the PBO problem that satisfies Ψ while minimizing $\mathcal{F}$.

A PBO-based SAT instance can be solved using modern PBO solvers that even support multiple objective functions. Several PBO-based approaches like (26–28) have already employed these solvers for the optimization of different testing aspects successfully.

3.3. Integer linear programming

Integer linear programming has been extensively used for solving many practical problems such as the traveling salesman, the matching problem, the quadratic assignment or the set covering problem (29). ILP defines an optimization problem with linear objective function, subject to a set of linear constraints that include integer variables:

$$\begin{aligned} &\text{minimize / maximize } \mathcal{F}^T x \\ &\text{subject to } Ax \leq b \text{ with } x \in \mathbb{W} \end{aligned} \quad (3)$$

Where x is a vector of unknown values referred to as decision variables that are restricted to the set of integers. $A = [a_{ij}]_{m \times n}$ is generally a matrix with $a_{ij} \in \mathbb{R}$ and creates the constraints bounded to the $b \in \mathbb{R}^m$. An ILP problem in which the integer variables are restricted to be 0 or 1 is called a 0 – 1 ILP problem. Example 3 shows the solution to an exemplary ILP instance.

Example 3. The optimal solution of $\max\{x + y\}$ subject to $-x + y \geq 1$, $-8x + 9y \leq 10$ with $(x, y) \in \mathbb{W}$ is $(x, y) = (1, 2)$. This solution both satisfies the inequalities and provides the maximum value for the objective function, as shown in Fig. 2.

Once the problem is formulated as an ILP model, different highly specialized ILP solvers can be invoked to efficiently solve it to optimality. The next section describes how the proposed methods exploit the PBO and ILP techniques to formulate the instrument access based on the extracted model of the IJTAG network.

4. Proposed test scheduling optimization

This section proposes two optimization methods for performing the test scheduling in IJTAG networks while considering multiple power domains. By this, it is ensured that the resulting test set contains only power-safe tests. More precisely, the first proposed technique of this

¹ For the sake of a comprehensive presentation, the CSU signals are not shown in Fig. 1.

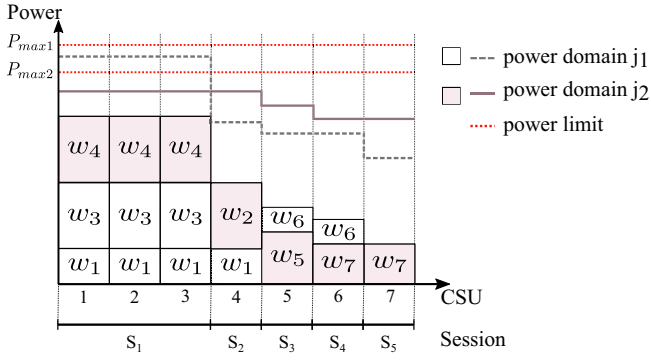


Fig. 3. Scheduling optimization problem.

work exploits an incremental PBO-based approach for minimizing the instrument scheduling time by determining local optimal test sessions successively. On top of this, an enhanced approach has been developed by taking advantage of ILP-based techniques to determine a global optimal test schedule – without the need for any incremental calculations. The proposed PBO method allows calculating the resulting test schedule in relatively low run-time and, despite the local optimality, the results show less than 10% mean absolute percentage error (MAPE) in comparison to the global optima.

4.1. PBO-based scheduling model

Both proposed test scheduling methods are applicable on arbitrary IJTAG networks comprising SIBs, ScanMuxes, SCBs, TDRs, and branching nodes. Every instrument requires a number of accesses that has to be fulfilled during the test process. The power consumption of each instrument, the number of power domains, and their power limits are considered as the input data to the scheduler. The inclusion or exclusion of instruments in/from a specific test session is well suitable for being mapped to the SAT problem, as done in this work. On top of this, the Boolean model is enriched by PB aspects to reflect the integer constraints regarding the power and number of accesses as well. Thus, the evaluation of this pseudo-Boolean function allows assessing these criteria of a determined test schedule. Finally, for not just assessing a test schedule but for determining a (local) optimal one, a PBO function $\Psi : \mathbb{B}^n \rightarrow \mathbb{R}$ has been introduced, which allows optimizing a test schedule while considering the given power and access constraints.

The proposed PBO-based scheduling approach can be divided into the following four main steps:

1. Extracting the **Boolean representation** of the network from the ICL file. More precisely, three main IJTAG building blocks have been developed allowing for generating the overall SAT instance out of a given ICL description.
2. Introducing the **power constraints** as PB components with respect to the given instruments' power consumption and the predefined power domain limits.
3. Adding the **objective function** to maximize the number of concurrently active instruments in every test session.
4. The **incremental invocation** of the SAT solver to provide an optimized access to the available instruments and subsequent masking of the fully covered instruments of currently optimized session to exclude them from the subsequent invocations.

An overview of the proposed scheduling scheme is depicted in Fig. 3. It is assumed that seven instruments w_1 to w_7 which are shown as boxes, are required to be tested during the minimum possible time. This is achieved by organizing the boxes over a minimized number of strips which represent the CSUs. The instruments of Fig. 3 are divided into two power domains indicated with different colors with the power limits

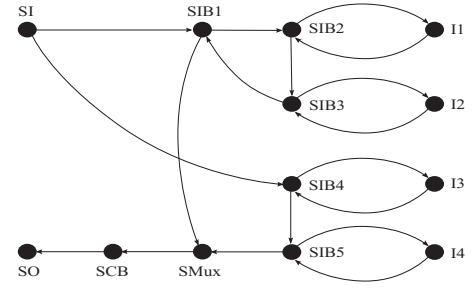


Fig. 4. The graph model of the IJTAG network of Fig. 1.

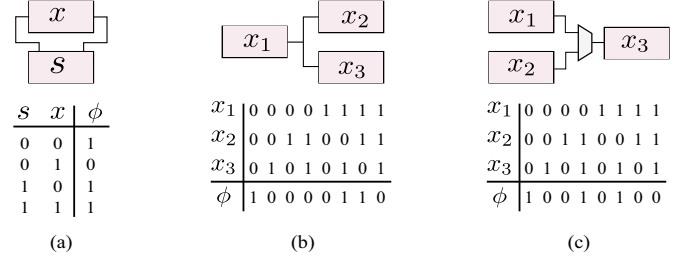


Fig. 5. Structures used for IJTAG network modeling.

P_{max1} and P_{max2} . The number of boxes equals the number of required access to the corresponding instrument. For example, three boxes of w_4 in Fig. 3 indicate three given accesses in the test pattern. The height of each box represents its power consumption. The horizontal axis is divided into CSU units. A set of successive CSUs, that contains the same selection of instruments, is called a (test) session, as indicated by S_1 to S_5 in Fig. 3. For instance, the first three successive strips contain the same set of boxes forming the first session. The control signals are applied through the control circuitry during each CSU. These control signals are omitted from the Figures for a cleaner presentation. The corresponding control signals are created by the IJTAG-embedding EDA tool and driven by a controller. Thus, no dependencies between these signals exist over different sessions and CSUs. Every session in this scheme represents a scan chain that can be reordered due to their independence. The last CSU of every session contains the SIB configuration data required for creating the next scan chain. Over the scheduled strips, the sum of instruments' power consumption does not violate the corresponding power limit. This is shown with dashed and solid lines for domains j_1 and j_2 , respectively. The power consumption of the scan chain elements are not considered since they are negligible compared with instruments' power consumption. The proposed scheduler tries to allocate the boxes to the strips such that the number of introduced CSUs are minimized.

4.1.1. Boolean representation of IJTAG networks

Every complex IJTAG network is constructed of several constitutive elements connected in one of the three different types, namely branches, merging points, and parallel connections. A complete network model is realized by combining the CNF model of these three building blocks, which can be identified by analyzing the network graph, as introduced in (14). The process starts by reading the ICL netlist and extracting a graph in which every path represents a scan chain starting from SI and ending at SO. This is assumed to be a directed acyclic graph as the IEEE 1687 Std. deprecates the creation of loops, for instance, by classifying the loop-containing SIB-design as a “bad SIB topology”. After classifying the network structures, the resulting CNF is added to the PBO instance as a constraint. Fig. 4 shows the graph representation of the network, as given in Fig. 1. Despite the graph model of work (14), SIBs are defined with one node in order to reduce the number of clauses in the resulting CNF expression. A Boolean variable is assigned to each node of the

graph. A logical value 0 implies that the corresponding node is not included in the active chain and, therefore, is not accessible. A variable with value 1 indicates that the node is located on an active chain. The CNF model of the network defines all potential scan chains between SI and SO. Fig. 5 shows the main building blocks and their truth table that are used to model the structure of IJTAG networks.

4.1.1.1. Modeling the SIBs. Fig. 5.a shows the SIBs hosting the sub-network x that can be of any type of network element. On the client side, a SIB works as a switch enabling the access to a hierarchy. Thus, as long as it is not activated, the sub-network is not accessible. On the other hand, an activated SIB can decide on switching the hierarchy ON or OFF. This can be expressed as the following CNF clause:

$$\Phi = s \vee \neg x \quad (4)$$

The connections between TDRs and instruments follow the same model since they are connected in parallel and data transfer is controllable by control signals. In case of disabling the instruments, TDRs remain connected to the network to ensure the flow of data to other elements. In addition to the host ports, SIBs have two other ports for connecting the rest of the network as normal elements, whose behavior is modeled by the other building blocks.

4.1.1.2. Modeling the branches. Fig. 5.b is a generalized model for plain chains and branches that are used in (19) for retargeting purposes. Plain chains can be considered as a special case of branches in which the source element has only one successor. The Boolean model of a branch should characterize three main features that are summarized in the truth table of Fig. 5.b. Firstly, only one active chain can exist at a time since the IEEE 1687 Std. allows only one serial path between SI and SO during a CSU. This requirement is satisfied by invalidating the states in which both x_2 and x_3 are set to True. This is achieved by the clause

$$(\neg x_2 \vee \neg x_3). \quad (5)$$

The second feature concerns the state that the source node x_1 is being activated. In this case, one of the branches has to be included on the current chain to provide the data flow to the next level. The related clause is given by

$$(\neg x_1 \vee x_2 \vee x_3). \quad (6)$$

The last feature addresses the activation of one of the successors in a branching node. In such states, the data has to be provided by a predecessor, which is the source node x_1 here. The CNF clause associated with this feature is

$$(x_1 \vee \neg x_2) \wedge (x_1 \vee \neg x_3). \quad (7)$$

Finally, the generalized CNF representation of a branching node with $n - 1$ successors is the conjunction of the extracted Boolean clauses as follows:

$$\Phi = (\neg x_1 \vee x_2 \vee \dots \vee x_n) \wedge \bigwedge_{i,j \in \{2..n\} \& i \neq j} (\neg x_i \vee \neg x_j) \wedge \bigwedge_{i=2}^n (x_1 \vee \neg x_i) \quad (8)$$

In the special case with $n=2$, only one successor exists, this circumstance creates in conjunction with the source node a so called plain chain. Such a plain chain does not include any branching or merging points and all elements on such a chain have the same activation condition: either all of them are accessible or none of them are being activated. The defining CNF for two adjacent nodes on plain chain is reduced to a bi-implication:

$$\Phi = x_1 \leftrightarrow x_2 = (x_1 \vee \neg x_2) \wedge (\neg x_1 \vee x_2) \quad (9)$$

4.1.1.3. Modeling the merging nodes. The last building blocks constituting the IJTAG network are merging points. Contrary to the branches, here, several paths meet and lead to one output successor. This typically applies where the ScanMuxes are inserted. The truth table of Fig. 5.

d shows the different possible assignments to the elements of a merging node with two predecessors. Two inputs of the multiplexer cannot be activated simultaneously and, hence, this circumstance is reflected by the first clause of the CNF. Furthermore, the selection of either of the inputs results in the activation of the output. As the last feature, when the output is on an active chain, one of the predecessor nodes has to be on the same chain to ensure the flow of data. These three aspects are satisfied by the following Boolean expression:

$$\Phi = (\neg x_1 \vee \neg x_2) \wedge (\neg x_2 \vee x_3) \wedge (\neg x_1 \vee x_3) \wedge (x_1 \vee x_2 \vee \neg x_3) \quad (10)$$

The generalized CNF representation for a merging node with $n - 1$ predecessors is expressed as follows:

$$1.0! \Phi = (x_1 \vee x_2 \vee \dots \vee \neg x_n) \wedge \bigwedge_{i,j \in \{1..n-1\} \& i \neq j} (\neg x_i \vee \neg x_j) \wedge \bigwedge_{i=1}^{n-1} (x_i \vee \neg x_n) \quad (11)$$

The overall Boolean network model is progressively constructed by conjugating the newly extracted CNF clauses to the SAT instance.

4.1.2. Power constraints

The power domain constraint implies that the sum of the power consumption of the concurrently activated instruments on the current scan chain during a CSU should not exceed the power limit of the power domain:

$$\sum_{p_i \in CSU_n} p_i \leq P_{max} \quad (12)$$

In a multi-power domain network, this criterion applies to all power domains. Assuming that the instruments of the network are powered by j different power domains while each of them has the following attributes: The maximum power limit $P_{max, j}$ and the power constraint $\mathcal{E}_j$ that is defined by adding the power consumption p_i of each instrument as a PB co-factor to the corresponding Boolean variable x_i such that activation of the instrument introduces its power consumption to be considered in $\mathcal{E}_j$. Consequently, the following overall power constraint has to be satisfied by the valid solution:

$$\bigwedge_{j \in \mathcal{J}} \left(\sum_{i \in \mathcal{I}_j} p_i \cdot x_i \right) \leq P_{max, j} \quad (13)$$

where $\mathcal{J}$ is the set of accessible instruments powered by domain j .

4.1.3. Objective function

The PB objective function $\mathcal{F}$ is defined over every session and maximizes the allocated power consumption in every strip by piling up the available instruments in the earliest stages so that the number of achieved CSUs are minimized. Assuming that $I_i(a_i, p_i)$ is the i -th instrument with a_i number of accesses and power consumption p_i , the objective function is implemented by accumulating the Boolean variables x_i that are weighted by PB co-factors p_i :

$$\mathcal{F}(x_1, \dots, x_N) = \sum_{i=1}^N (p_i \cdot x_i) \quad (14)$$

The already covered instruments are masked from the next iterations by setting the corresponding variables to 0.

4.1.4. Incremental invocation

The proposed scheduler follows the progressive optimization approach described in Algorithm 1 over the successive test sessions. In Fig. 3, it is assumed that the CSU strips are initially empty and all instrument boxes are available in a pool. The first set of instruments are allocated to the first CSU such that the maximum power consumption is obtained while considering the concurrent accessibility of the elements. This is the case about w_1 , w_3 , and w_4 . This selection remains unchanged over the first three CSUs creating the first test session. By completing the access to w_3 and w_4 , the session is closed, and these instruments are

excluded from the future schedules by deactivating their corresponding literals. On the other hand, the remaining access required by w_1 is considered in the subsequent iterations. By this, in every run of the PBO solver, the optimization problem is updated with new instrument access requirements. This process continues until the instrument access pool is empty and, hence, no further elements have to be scheduled. The access to at least one instrument is completely fulfilled in every session and, hence, the next session has to process a reduced instrument pool only. Therefore, by assuming N as the number of instruments, this algorithm provides the upper bound N for the maximum number of iterations. Accordingly, the largest search space is explored in the first session with 2^N Boolean decisions. The search space is constrained by power constraints and further bounded by a minimum number of N Boolean clauses of the CNF definition.

Algorithm 1. PBO-based scheduling.

Input: ICL , $constraints$
Output: *Minimized instrument access schedule*

```

1  $instsToSchedule$  = Set of all instruments that are not
  scheduled yet;
2 while ( $instsToSchedule \neq \emptyset$ ) do
3   Create new session;
4   Optimize( $instsToSchedule$ ,  $constraints$ );
5   if ( $Instrument[i]$  has the minimum access) then
6     Remove  $Instrument[i]$  from
       $instsToSchedule$ ;
7     Update the remaining access for members
      of  $instsToSchedule$ ;
8   end
9 end
10 function Optimize( $instsToSchedule$ ,  $Constraints$ ):
11   Collect the highest possible sum of power
    consumption while considering the constraints.
12 return
```

4.2. ILP-based optimization model

In the previously proposed PBO-based method, the PBO solver decides on the value of pure Boolean decision variables yielding a rather fast solving process. However, to create a global optimization scheme for the test scheduling, new integer decision variables are introduced to decide on the number of CSUs in each test session. This circumstance forms an integer linear programming problem that can be solved over a single run of an ILP solver, as described in Algorithm 2. The computational cost of this algorithm depends on three main factors. Assuming N as the number of instruments, the objective function will have N integer variables. Considering the Boolean variables for instruments, the solver determines the value of a total of $N^2 + N$ decision variables. This search space is constrained by $N \cdot D$ equations defining the power constraint where D represents the number of power domains. The next factor that heavily constrains the search space concerns the inequalities that define the structural constraint of the given network. The number of inequalities grows linearly with the number of edges in the network's graph.

Algorithm 2. ILP-based scheduling.

Input: ICL , *power limits*, *access requirements*
Output: *Minimized instrument access schedule*

```

1  $N$  = Number of instruments
2  $D$  = Number of power domains
3  $objectiveFunction$  =  $\min \sum CSUs$ 
4 for ( $i=0$  to  $N-1$ ) do
5   for ( $d=1$  to  $D$ ) do
6      $powerConstraints$   $\leftarrow$  Create power
      constraints
7   end
8 end
9 for ( $i=0$  to  $N-1$ ) do
10    $accessConstraints$   $\leftarrow$  Create access constraints
11 end
12 callSolver( $objectiveFunction \cup powerConstraints \cup$ 
   $accessConstraints$ )
13 function callSolver( $ilpProblem$ ):
14   Solve the objective function subjected to power
    and access constraints.
15 return
```

4.2.1. Objective function

In the global optimization approach, the goal of the optimization is minimizing the overall number of required CSUs to cover all instruments. However, instead of formulating multiple local objective functions, only one objective function is created for all possible test sessions. Therefore, new Boolean decision variables are introduced to every hypothetical test session. In the worst-case scenario, every instrument is accessed separately through an individual scan chain. Thus, in an IJTAG network with n instruments, a maximum of n sessions for the entire scheduling process is expected. Considering this value as an upper bound, the decision matrix for such a network is represented as follows:

$$M = \begin{matrix} & \text{inst.1} & \text{inst.2} & \dots & \text{inst.n} & \\ \begin{matrix} \text{session 1} \\ \vdots \\ \text{session n} \end{matrix} & \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1n} \\ \vdots & & & \\ x_{n1} & x_{n2} & \dots & x_{nn} \end{bmatrix} & \end{matrix}$$

Each row of matrix M represents a test session over which the Boolean decision variable x_{ij} indicates the inclusion of the corresponding instrument or its exclusion from the i -th scan chain according to the solver's decision. On the other hand, every column of M reflects the engagement of the related instrument over different scan configurations in the final optimized solution. All the elements of column j have an identical number of given accesses, power consumption, and power domain, namely A_j , p_j , and d_j . In other words, all elements of a column refer to the same instrument, although on different sessions. The identical number of CSUs is applied to all active instruments of a row which provides them with the same number of accesses. Accordingly, an integer decision variable a_i representing the number of fulfilled accesses is introduced to row i of matrix M . As the scheduler tries to cover all instruments via the minimum number of CSUs or accesses, the objective function is defined as follows:

$$\text{Min} \sum_{i=1}^n a_i \quad (15)$$

This objective function guarantees that the ILP solver optimizes the instrument access scheduling to provide a minimized overall test time.

4.2.2. ILP constraints

Three constraints described below outline the optimization environment. The first one addresses the network structure and restricts the search space to concurrently accessible elements. The next constraint considers the power limit of different domains, and the last one fulfills the number of required access for each instrument.

Structural constraints investigate whether the concurrently activated elements can create a path between SI and SO. The CNF representation, as already discussed in Section 4.1.1, is used to generate the equivalent ILP constraint by transforming every CNF clause to an ILP inequality. Example 4 shows this process for two successive elements x_1 and x_2 .

Example 4.

$$\Phi = (x_1 \vee \neg x_2) \wedge (\neg x_1 \vee x_2) \Rightarrow \begin{aligned} x_1 - x_2 &\geq 0 \\ -x_1 + x_2 &\geq 0 \end{aligned}$$

The created inequalities imply that both Boolean variables x_1 and x_2 should have the same value and this correctly mimics the behavior described by the CNF.

Power constraints are defined in the same way as done for their counterparts in PBO method. Assuming p_{ij} as the power consumption of the j -th instrument of the i -th row of the decision matrix M , and $P_{max,k}$ as the power limit of the k -th power domain, the following power constraint applies to every row of M :

$$\sum_{j \in k} p_{ij} \cdot x_{ij} \leq P_{max,k} \text{ with } 1 \leq k \leq K \quad 1 \leq i \leq n \quad (16)$$

This guarantees that the overall power consumption of active instruments on a chain does not exceed the related power domain's limit. When considering all rows of the decision matrix, the total number of created power constraints equals $n \times k$.

Access constraints ensure the complete access coverage for every instruments over the different sessions. Every access is performed over one CSU. Assuming that a_i represents the number of access to the instruments in i -th session, and A_j is the required access of j -th instrument in decision matrix, the access constraint is defined as

$$\sum_{i=1}^n a_i \cdot x_{ij} = A_j \text{ with } 1 \leq j \leq n \quad a_i, A_j \in \mathbb{W}. \quad (17)$$

Since both a_i and x_{ij} are decision variables, this results in a non-linear constraint that cannot be resolved directly through linear algorithms. Therefore, the Glover's method proposed in (30) is used to linearize this equation by considering $y_{ij} = x_{ij} \cdot a_i$ and adding the following assumptions as alternative linear constraints:

$$\begin{aligned} \sum_{i=1}^n y_{ij} &= A_j \text{ with } 1 \leq j \leq n, y_{ij}, A_j \in \mathbb{W} \\ 0 &\leq y_{ij} \leq A x_{ij}, a_i - A(1 - x_{ij}) \leq y_{ij} \leq a_i \end{aligned} \quad (18)$$

Where A is the maximum number of given accesses among all instruments. These inequalities represent an $n \times n$ matrix $\bar{M}$ with y_{ij} as its elements:

$$\bar{M} = \begin{bmatrix} y_{11} & \dots & y_{1n} \\ \vdots & \ddots & \vdots \\ y_{n1} & \dots & y_{nn} \end{bmatrix}.$$

Depending on the value of x_{ij} , the value of y_{ij} is either 0 or a_i .

The described constraints define an ILP optimization environment over which the available powerful ILP solvers are able to solve the instrument scheduling problem using the proposed objective function.

5. Experimental results

This section presents the experimental evaluation of the proposed test schedulers which orchestrate PBO-based and ILP-based techniques

Table 1

IJTAG Benchmark networks' structural modeling results.

Benchmark network	#(nodes, edges)	#instruments	#variables	#clauses
Class	Name			
B	Mingle	(27, 39)	8	35
	BasicSCB	(33, 42)	5	39
	TreeFlat	(38, 61)	11	38
C	q12710	(52, 78)	23	75
	a586710	(93, 124)	22	115
	t512505	(289, 447)	128	417
	p22810	(520, 789)	242	762
	p34392	(284, 380)	73	357
	p93791	(1763, 2358)	550	2313
	TrapOrFlap	(42, 61)	12	45
A	N17D3	(52, 66)	27	79
	N32D6	(79, 101)	44	123
	N73D14	(155, 200)	90	245
	N132D4	(293, 371)	172	465
	NE600P150	(1597, 1997)	793	2183
	NE1200P430	(3253, 4063)	1629	4502
				8202

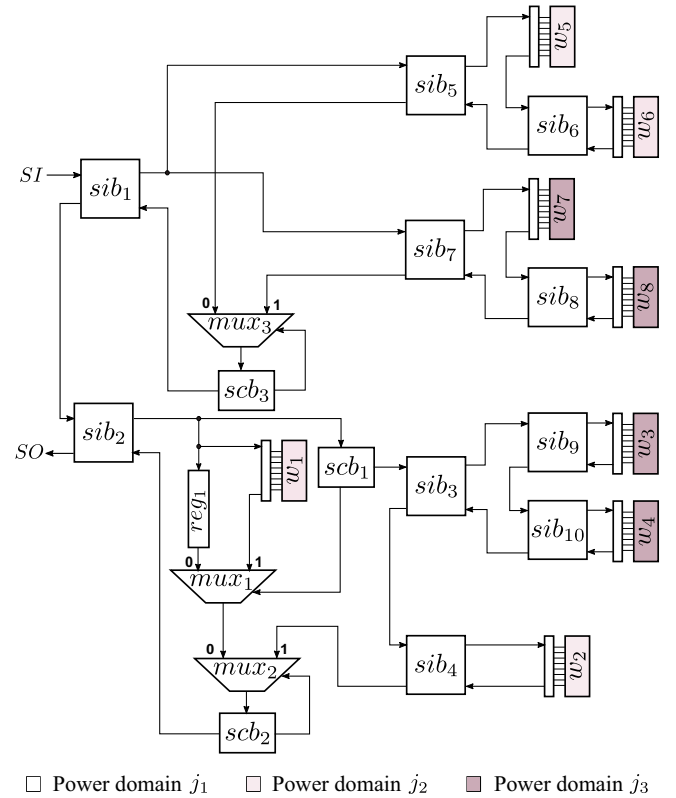


Table 2
Scheduling scenario for Mingle network.

Instrument	Domain	Power	#accesses
w_1	2	85	32
w_2	2	115	18
w_3	3	93	20
w_4	3	110	12
w_5	2	76	10
w_6	2	134	25
w_7	3	125	35
w_8	3	100	13

of basic, classic, and advanced, which are indicated by 'B', 'C', and 'A' in Table 1. Despite being relatively small as is shown in Fig. 6, the Mingle network is a complex instance of the ITC'16 benchmark set since it contains all structural features like branching and merging points, SIBs, SCBs, and parallel branches. Consequently, the result of test scheduling for this network is presented in detail. The other networks follow the same procedure and a summary of their results are reported in Table 4.

Table 2 shows the scenario created for the evaluation of the test scheduling for the Mingle network. The whole network is divided into three power domains indicated with different colors in Fig. 6. The eight wrapped instruments w_1 to w_8 are allocated to power domains j_2 and j_3 with a power limit 140. For the sake of a more comprehensible presentation, the limit for the power domain j_1 is not taken into account. The number of required access for each instrument and their power consumption has been randomly generated by the framework since they are not provided in ICL netlist. Note that these values can be adopted from any arbitrary power simulation or analyzing tool.

According to the scenario of Table 2, the proposed framework scheduled a total of 165 instrument accesses in 90 CSUs without violating the power domains' limits. In the incremental PBO method, an extra option of successive access scheduling can be introduced, as presented in Fig. 7. That is, the solver is conducted to schedule the instruments such that they are accessed in successive sessions. Albeit re-ordering the sessions in a non-successive schedule is allowed which

typically results in shorter overall access time.

The experimental evaluation of non-successive scheduling for the Mingle network with the same power configuration is given in Table 3 comparing the number of required CSUs using PBO and ILP methods. All considered scenarios with randomly generated accesses in Table 3 show the capability of the PBO method to estimate the optimal overall test time in CSUs. Accordingly, for evaluating the efficiency of the proposed method, different scenarios with a random number of access and power consumption, as shown in Table 4, have been applied to other benchmark networks. Each network is divided into three power domains. The sum of power consumption for each domain and their power limits are given in the table. Due to the page limitation, the column $\sum$ accesses shows the sum of required accesses for each network instead of mentioning every number of accesses separately. The number of CSUs over which all instruments can be covered is reported for both PBO and ILP methods in the next two columns. Comparing these columns shows a promising estimation of the global optimal value by the PBO-scheduler in a short CPU run-time which is reported in the rightmost columns for both methods.

Despite providing the optimal solution, the run-time of the ILP method is considerably higher than the one of the PBO-based approach. Consequently, for the networks with more than 800 CNF clauses, it takes several hours to solve the problem. These cases are indicated by ∞ in the last column. However, since the real integrated circuits use several test channels instead of one large JTAG network, the use of the ILP scheduler can still be extended to larger circuits. On the other hand, the experimental results clearly prove that the PBO-based test scheduler is capable of processing even large JTAG network structures with multi-power domains representing industrial-relevant benchmark candidates. Besides this, the proposed PBO scheduler allows for determining highly effective but power-safe test schedules.

6. Conclusion

This paper proposed two novel PBO-based and ILP-based test scheduling optimization schemes for multi-power domain JTAG

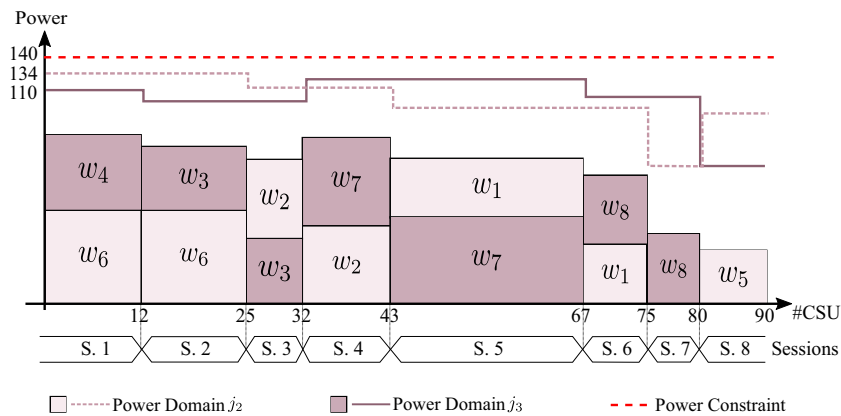


Fig. 7. Result of the successive test scheduling for Mingle network.

Table 3
Test scheduling in Mingle network.

Scenario	#accesses								#CSUs	
	w_1	w_2	w_3	w_4	w_5	w_6	w_7	w_8	PBO	ILP
1	27	12	33	22	18	29	40	19	114	114
2	28	36	42	19	22	17	10	50	123	121
3	34	53	42	29	37	19	36	58	165	165
4	43	71	117	58	52	90	48	53	289	276
5	119	227	175	163	284	111	324	98	817	817
6	342	294	460	611	370	394	436	229	1736	1736

Table 4

Results of non-successive test scheduling for ITC'16 benchmark networks using PBO and ILP methods.

Network	Power domain			$\sum$ accesses	#CSU		#chains		run-time [s]	
	ID	$\sum$ power	Limit		PBO	ILP	PBO	ILP	PBO	ILP
Mingle	1	27	20	200	114	114	8	8	0.019	0.73
	2	410	140							
	3	428	140							
BasicSCB	1	34	30	88	52	52	5	5	0.012	0.52
	2	328	180							
	3	473	180							
TreeFlat	1	13	30	158	30	26	8	10	0.020	42.0
	2	530	330							
	3	412	290							
TrapOrFlap	1	27	40	153	49	46	11	12	0.05	5.70
	2	521	190							
	3	490	180							
q12710	1	52	55	377	191	191	21	22	0.094	1.14
	2	1177	160							
	3	1274	170							
a586710	1	65	70	279	54	45	19	16	0.23	6.52
	2	771	265							
	3	1281	385							
t512505	1	130	30	1684	476	427	110	92	3.215	1685.21
	2	4458	190							
	3	5040	180							
p22810	1	244	30	3090	805	–	214	–	12.654	∞
	2	9110	190							
	3	8974	180							
p34392	1	350	90	953	96	96	48	21	3.41	824.33
	2	2742	400							
	3	2327	450							
p93791	1	552	600	7122	255	–	231	–	180.54	∞
	2	2169	1200							
	3	20,388	1100							
N17D3	1	53	55	488	174	173	27	24	0.087	1803.53
	2	1378	200							
	3	1066	150							
N32D6	1	79	100	655	119	117	35	37	0.223	1801.13
	2	1478	200							
	3	1812	250							
N73D14	1	94	100	900	120	120	12	12	0.843	1309.42
	2	3444	300							
	3	3468	300							
N132D4	1	367	250	2453	361	–	137	–	5.008	∞
	2	7059	300							
	3	6487	300							
NE600P150	1	795	500	10,524	188	–	184	–	159.622	∞
	2	29,101	2200							
	3	29,441	2300							
NE1200P430	1	1631	700	21,187	282	–	274	–	555.825	∞
	2	63,040	3100							
	3	59,358	3300							

networks that are seamlessly integrated into a fully-automated power-aware test scheduling framework. More precisely, in the first step, a seamless SAT model for complex IJTAG network structures had been designed, which was next exploited to establish two optimization approaches based on pseudo-Boolean and ILP optimization techniques. The entire framework had been implemented in C++ that support the IJTAG's inter-connectivity language. The PBO-based scheduler uses the *clasp* solver to provide a local optimal solution for even large networks with over one thousand instruments forming industrial representative benchmark candidates. On the other hand, the ILP-based scheduler uses the *CPLEX* ILP solver yielding a global optimal that is practical for small to middle-sized networks and was employed as an evaluation means for the performance of the PBO-based scheduler. In the end, both schedulers allow for determining highly effective but power-safe tests and, by this, contributes to reduced test costs when testing complex and power-critical SoC designs.

Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

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Payam Habiby received his B.S. and M.S. degrees in electrical engineering from the University of Guilan, Rasht in 2007 and 2012 respectively. After receiving his master's degree, he was with NIOC as an electronics engineer until 2019. He is currently a researcher at the German Research Center for Artificial Intelligence (DFKI) and a Ph.D. candidate at the University of Bremen in the Group of Computer Architecture with Prof. Drechsler. His research interests include Design for Testability, embedded test & repair, and formal techniques of test optimization.



Sebastian Huhn received the B.S. and M.S. degrees and the Dr.-Ing. doctorate degrees in computer engineering from the University of Bremen, Bremen, Germany, in 2012, 2014, and 2020, respectively. He is currently a PostDoc in the Group of Computer Architecture at this university and a Senior Researcher at the German Research Center for Artificial Intelligence, Bremen. His current research interests include design for test, reliability, and debug, in particular, novel test generation, test interfaces and pattern retargeting techniques. Besides this, he is interested in formal techniques, formal reliability analysis & enhancement of complex systems and circuits.



Rolf Drechsler received the Diploma and Dr. phil. nat. degrees in computer science from the JohannWolfgang Goethe University in Frankfurt am Main, Germany, in 1992 and 1995, respectively. He worked at the Institute of Computer Science, Albert-Ludwigs University, Freiburg im Breisgau, Germany, from 1995 to 2000, and at the Corporate Technology Department, Siemens AG, Munich, Germany, from 2000 to 2001. Since October 2001, Rolf Drechsler is Full Professor and Head of the Group of Computer Architecture, Institute of Computer Science, at the University of Bremen, Germany. In 2011, he additionally became the Director of the Cyber-Physical Systems Group at the German Research Center for Artificial Intelligence (DFKI) in Bremen. His current research interests include the

development and design of data structures and algorithms with a focus on circuit and system design. He is an IEEE Fellow.